



1. Description

1.1. Project

Project Name	BootloaderCode
Board Name	custom
Generated with:	STM32CubeMX 6.11.1
Date	04/30/2026

1.2. MCU

MCU Series	STM32L4
MCU Line	STM32L4x6
MCU name	STM32L496ZGTx
MCU Package	LQFP144
MCU Pin number	144

1.3. Core(s) information

Core(s)	Arm Cortex-M4
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3. Pins Configuration

Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	PE2	I/O	GPIO_EXTI2	ABORT_STATUS
3	PE4	I/O	GPIO_EXTI4	READY_INIT_STATUS
5	PE6	I/O	GPIO_EXTI6	PREP_STATUS
6	VBAT	Power		
10	PF0 *	I/O	GPIO_Output	DAC1_Trigger
11	PF1 *	I/O	GPIO_Output	DAC2_Trigger
12	PF2 *	I/O	GPIO_Output	DAC1_RESET
13	PF3 *	I/O	GPIO_Output	DAC2_RESET
14	PF4 *	I/O	GPIO_Output	SEG3_2
15	PF5 *	I/O	GPIO_Output	SEG3_3
16	VSS	Power		
17	VDD	Power		
18	PF6 *	I/O	GPIO_Output	DIS_CH1_OUT
19	PF7 *	I/O	GPIO_Output	DIS_CH2_OUT
20	PF8 *	I/O	GPIO_Output	DIS_CH3_OUT
21	PF9 *	I/O	GPIO_Output	DIS_CH4_OUT
22	PF10 *	I/O	GPIO_Output	DIS_CH5_OUT
23	PH0-OSC_IN (PH0)	I/O	RCC_OSC_IN	
24	PH1-OSC_OUT (PH1)	I/O	RCC_OSC_OUT	
25	NRST	Reset		
26	PC0	I/O	ADC1_IN1	
29	PC3 *	I/O	GPIO_Output	RS422_2_DE_RE
30	VSSA	Power		
31	VREF-	Power		
33	VDDA	Power		
34	PA0 *	I/O	GPIO_Output	RS422_LOOP_IN4
35	PA1 *	I/O	GPIO_Output	RS422_LOOP_IN3
36	PA2	I/O	USART2_TX	
37	PA3	I/O	USART2_RX	
38	VSS	Power		
39	VDD	Power		
40	PA4 *	I/O	GPIO_Output	SPI1_CS1
41	PA5	I/O	SPI1_SCK	
42	PA6	I/O	SPI1_MISO	
43	PA7	I/O	SPI1_MOSI	
44	PC4 *	I/O	GPIO_Output	SPI1_CS2

Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
46	PB0 *	I/O	GPIO_Output	SEG3_LE
47	PB1 *	I/O	GPIO_Output	SEG2_BC
48	PB2 *	I/O	GPIO_Output	SEG1_BC
49	PF11 *	I/O	GPIO_Output	ROTRY_ADD0
50	PF12 *	I/O	GPIO_Output	ROTRY_ADD1
51	VSS	Power		
52	VDD	Power		
53	PF13 *	I/O	GPIO_Output	ROTRY_ADD2
54	PF14 *	I/O	GPIO_Output	ROTRY_OE1
55	PF15 *	I/O	GPIO_Output	ROTRY_OE2
56	PG0 *	I/O	GPIO_Output	ROTRY_OE3
57	PG1 *	I/O	GPIO_Output	ROTRY_OE4
58	PE7 *	I/O	GPIO_Output	SEG1_1
59	PE8 *	I/O	GPIO_Output	SEG1_2
60	PE9 *	I/O	GPIO_Output	SEG1_3
61	VSS	Power		
62	VDD	Power		
63	PE10 *	I/O	GPIO_Output	SEG1_4
64	PE11 *	I/O	GPIO_Output	SEG2_1
65	PE12 *	I/O	GPIO_Output	SEG2_2
66	PE13 *	I/O	GPIO_Output	SEG2_3
67	PE14 *	I/O	GPIO_Output	SEG2_4
68	PE15 *	I/O	GPIO_Output	SEG3_1
71	VSS	Power		
72	VDD	Power		
73	PB12	I/O	SPI2_NSS	
74	PB13	I/O	SPI2_SCK	
75	PB14	I/O	SPI2_MISO	
76	PB15	I/O	SPI2_MOSI	
78	PD9 *	I/O	GPIO_Output	WD_SET2
79	PD10 *	I/O	GPIO_Output	WD_SET1
80	PD11 *	I/O	GPIO_Output	WD_SET0
81	PD12	I/O	TIM4_CH1	
83	VSS	Power		
84	VDD	Power		
87	PG2 *	I/O	GPIO_Input	ROTRY_MUX_IN1
88	PG3 *	I/O	GPIO_Input	ROTRY_MUX_IN2
89	PG4 *	I/O	GPIO_Input	ROTRY_MUX_IN3
90	PG5 *	I/O	GPIO_Input	ROTRY_MUX_IN4

Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
91	PG6 *	I/O	GPIO_Output	SEG3_4
92	PG7 *	I/O	GPIO_Output	SEG1_LE
93	PG8 *	I/O	GPIO_Output	SEG2_LE
94	VSS	Power		
95	VDDIO2	Power		
96	PC6 *	I/O	GPIO_Output	READY_LED_IO2
97	PC7 *	I/O	GPIO_Output	ABORT_LED_IO2
99	PC9	I/O	TIM3_CH4	
100	PA8	I/O	TIM1_CH1	
103	PA11	I/O	USB_OTG_FS_DM	
104	PA12	I/O	USB_OTG_FS_DP	
105	PA13 (JTMS/SWDIO)	I/O	SYS_JTMS-SWDIO	
106	VDDUSB	Power		
107	VSS	Power		
108	VDD	Power		
109	PA14 (JTCK/SWCLK)	I/O	SYS_JTCK-SWCLK	
110	PA15 (JTDI) *	I/O	GPIO_Output	SPI3_CS
111	PC10	I/O	SPI3_SCK	
112	PC11	I/O	SPI3_MISO	
113	PC12	I/O	SPI3_MOSI	
119	PD5	I/O	GPIO_EXTI5	DISCRETE_CH3_STS
120	VSS	Power		
121	VDD	Power		
123	PD7 *	I/O	GPIO_Output	RS422_1_DE_RE
124	PG9	I/O	USART1_TX	
125	PG10	I/O	USART1_RX	
126	PG11 *	I/O	GPIO_Output	RS422_LOOP_IN1
127	PG12 *	I/O	GPIO_Output	RS422_LOOP_IN2
130	VSS	Power		
131	VDDIO2	Power		
133	PB3 (JTDO/TRACESWO)	I/O	TIM2_CH2	
134	PB4 (NJTRST) *	I/O	GPIO_Output	SEG3_BC
139	PB8 *	I/O	GPIO_Output	ABORT_LED_IO1
140	PB9 *	I/O	GPIO_Output	READY_LED_IO1
142	PE1 *	I/O	GPIO_Input	DIS_CH2_IN
143	VSS	Power		
144	VDD	Power		

* The pin is affected with an I/O function

5. Software Project

5.1. Project Settings

Name	Value
Project Name	BootloaderCode
Project Folder	D:\Venkat\MainProjects\U_07_Flight_Control_System_Test_UnitV2\14.Firmware\F
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_L4 V1.18.1
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0xC00

5.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	Yes
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	Yes
Enable Full Assert	Yes

5.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_SPI3_Init	SPI3
4	MX_TIM1_Init	TIM1
5	MX_USB_DEVICE_Init	USB_DEVICE
6	MX_TIM4_Init	TIM4
7	MX_USART1_UART_Init	USART1
8	MX_USART2_UART_Init	USART2
9	MX_ADC1_Init	ADC1
10	MX_SPI1_Init	SPI1
11	MX_SPI2_Init	SPI2

Rank	Function Name	Peripheral Instance Name
12	MX_TIM2_Init	TIM2
13	MX_TIM3_Init	TIM3
14	MX_TIM5_Init	TIM5
15	MX_IWDG_Init	IWDG

1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32L4
Line	STM32L4x6
MCU	STM32L496ZGTx
Datasheet	DS11585 Rev2

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

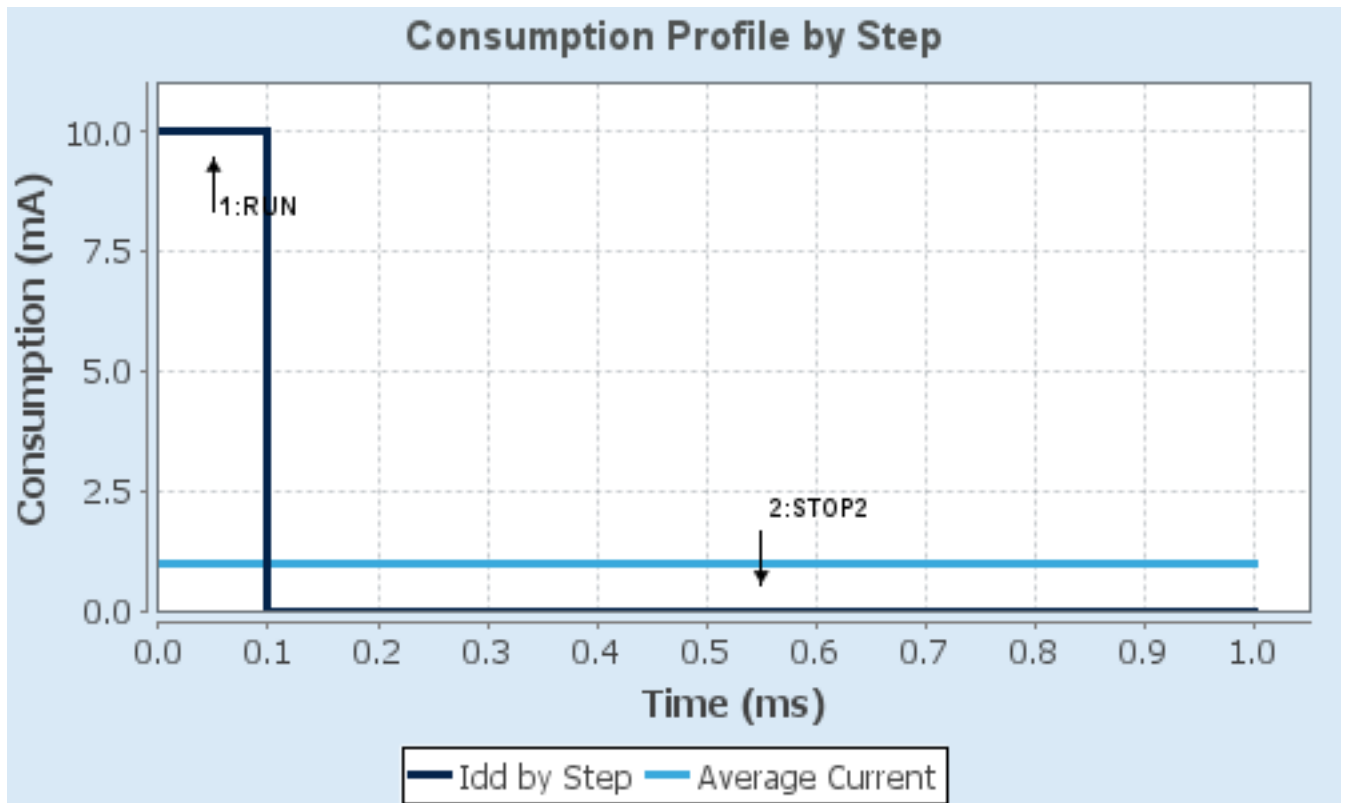
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP2
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-High	NoRange
Fetch Type	FLASH	n/a
CPU Frequency	80 MHz	0 Hz
Clock Configuration	HSE BYP PLL Flash-ON	ALL CLOCKS OFF
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	10 mA	2.69 μ A
Duration	0.1 ms	0.9 ms
DMIPS	100.0	0.0
Ta Max	104.04	105
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	1 mA
Battery Life	4 months, 19 days, 3 hours	Average DMIPS	100.0 DMIPS

1.6. Chart



2. Peripherals and Middlewares Configuration

2.1. ADC1

IN1: IN1 Single-ended

2.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler

Synchronous clock mode divided by 4 *

Resolution

ADC 12-bit resolution

Data Alignment

Right alignment

Scan Conversion Mode

Disabled

Continuous Conversion Mode

Enabled *

Discontinuous Conversion Mode

Disabled

DMA Continuous Requests

Disabled

End Of Conversion Selection

End of single conversion

Overrun behaviour

Overrun data preserved

Low Power Auto Wait

Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions

Enable

Enable Regular Oversampling

Disable

Number Of Conversion

1

External Trigger Conversion Source

Regular Conversion launched by software

External Trigger Conversion Edge

None

Rank

1

Channel

Channel 1

Sampling Time

2.5 Cycles

Offset Number

No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions

Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode

false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode

false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode

false

2.2. IWDG

mode: Activated

2.2.1. Parameter Settings:

Watchdog Clocking:

IWDG counter clock prescaler	256 *
IWDG window value	4095
IWDG down-counter reload value	2499 *

2.3. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

2.3.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Disabled
Data Cache	Enabled
Flash Latency(WS)	4 WS (5 CPU cycle)

RCC Parameters:

HSI Calibration Value	64
MSI Calibration Value	0
MSI Auto Calibration	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
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2.4. SPI1

Mode: Full-Duplex Master

2.4.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	16 Bits *
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	40.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Disabled *
NSS Signal Type	Software

2.5. SPI2

Mode: Full-Duplex Master

Hardware NSS Signal: Hardware NSS Output Signal

2.5.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	16 Bits *
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	8 *
Baud Rate	10.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Output Hardware

2.6. SPI3

Mode: Full-Duplex Master

2.6.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits *

First Bit	MSB First
Clock Parameters:	
Prescaler (for Baud Rate)	2
Baud Rate	40.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
Advanced Parameters:	
CRC Calculation	Disabled
NSSP Mode	Disabled *
NSS Signal Type	Software

2.7. SYS

Debug: Serial Wire

Timebase Source: SysTick

2.8. TIM1

Trigger Source: ITR0

Clock Source : Internal Clock

Channel1: PWM Generation CH1

2.8.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	8000-1 *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	100-1 *
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable
Slave Mode Controller	Slave mode disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	

- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

2.9. TIM2

Clock Source : Internal Clock

Channel2: PWM Generation CH2

2.9.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	8000-1 *
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	100-1 *
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (32 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

2.10. TIM3

Clock Source : Internal Clock

Channel4: PWM Generation CH4

2.10.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	8000-1 *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	100-1 *
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 4:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

2.11. TIM4

Clock Source : Internal Clock

Channel1: PWM Generation CH1

2.11.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	8000-1 *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	300-1 *
Internal Clock Division (CKD)	No Division
auto-reload preload	Enable *

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

2.12. TIM5

Trigger Source: ITR0

Clock Source : Internal Clock

2.12.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	800-1 *
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	100000 *
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable
Slave Mode Controller	Slave mode disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

2.13. USART1

Mode: Asynchronous

2.13.1. Parameter Settings:

Basic Parameters:

Baud Rate	57600 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

2.14. USART2

Mode: Asynchronous

2.14.1. Parameter Settings:

Basic Parameters:

Baud Rate	57600 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

2.15. USB_OTG_FS

Mode: Device_Only

2.15.1. Parameter Settings:

Speed	Full Speed 12MBit/s
Low power	Disabled
Battery charging	Disabled
Link Power Management	Disabled
Use dedicated end point 1 interrupt	Disabled
VBUS sensing	Disabled
Signal start of frame	Disabled

2.16. USB_DEVICE

Class For FS IP: Communication Device Class (Virtual Port Com)

2.16.1. Parameter Settings:

Basic Parameters:

USBD_MAX_NUM_INTERFACES (Maximum number of supported interfaces)	1
USBD_MAX_NUM_CONFIGURATION (Maximum number of supported configuration)	1
USBD_MAX_STR_DESC_SIZ (Maximum size for the string descriptors)	512
USBD_SELF_POWERED (Enabled self power)	Enabled
USBD_DEBUG_LEVEL (USBD Debug Level)	0: No debug message
USBD_LPM_ENABLED (Link Power Management)	1: Link Power Management supported

Class Parameters:

USB CDC Rx Buffer Size	2048
USB CDC Tx Buffer Size	2048

2.16.2. Device Descriptor:

Device Descriptor:

VID (Vendor Identifier)	1155
LANGID_STRING (Language Identifier)	English(United States)
MANUFACTURER_STRING (Manufacturer Identifier)	STMicroelectronics

Device Descriptor FS:

PID (Product Identifier)	22336
PRODUCT_STRING (Product Identifier)	STM32 Virtual ComPort
CONFIGURATION_STRING (Configuration Identifier)	CDC Config
INTERFACE_STRING (Interface Identifier)	CDC Interface

*** User modified value**

3. System Configuration

3.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PC0	ADC1_IN1	Analog mode	No pull-up and no pull-down	n/a	
RCC	PH0-OSC_IN (PH0)	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT (PH1)	RCC_OSC_OUT	n/a	n/a	n/a	
SPI1	PA5	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PA6	SPI1_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PA7	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
SPI2	PB12	SPI2_NSS	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PB13	SPI2_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PB14	SPI2_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PB15	SPI2_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
SPI3	PC10	SPI3_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PC11	SPI3_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PC12	SPI3_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
SYS	PA13 (JTMS/SWDIO)	SYS_JTMS-SWDIO	n/a	n/a	n/a	
	PA14 (JTCK/SWCLK)	SYS_JTCK-SWCLK	n/a	n/a	n/a	
TIM1	PA8	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM2	PB3 (JTDO/TRACESWIO)	TIM2_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	CESWO)					
TIM3	PC9	TIM3_CH4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM4	PD12	TIM4_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART1	PG9	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PG10	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
USART2	PA2	USART2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PA3	USART2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
USB_OTG_FS	PA11	USB_OTG_FS_DM	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PA12	USB_OTG_FS_DP	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
GPIO	PE2	GPIO_EXTI2	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	ABORT_STATUS
	PE4	GPIO_EXTI4	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	READY_INIT_STATUS
	PE6	GPIO_EXTI6	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	PREP_STATUS
	PF0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DAC1_Trigger
	PF1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DAC2_Trigger
	PF2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DAC1_RESET
	PF3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DAC2_RESET
	PF4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG3_2
	PF5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG3_3
	PF6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DIS_CH1_OUT
	PF7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DIS_CH2_OUT
	PF8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DIS_CH3_OUT
	PF9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DIS_CH4_OUT
	PF10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DIS_CH5_OUT
	PC3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RS422_2_DE_RE
	PA0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RS422_LOOP_IN4
	PA1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RS422_LOOP_IN3
	PA4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SPI1_CS1
	PC4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SPI1_CS2
	PB0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG3_LE
	PB1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG2_BC
	PB2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG1_BC

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PF11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ROTRY_ADD0
	PF12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ROTRY_ADD1
	PF13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ROTRY_ADD2
	PF14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ROTRY_OE1
	PF15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ROTRY_OE2
	PG0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ROTRY_OE3
	PG1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ROTRY_OE4
	PE7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG1_1
	PE8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG1_2
	PE9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG1_3
	PE10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG1_4
	PE11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG2_1
	PE12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG2_2
	PE13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG2_3
	PE14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG2_4
	PE15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG3_1
	PD9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	WD_SET2
	PD10	GPIO_Output	Output Push Pull	Pull-up *	Low	WD_SET1
	PD11	GPIO_Output	Output Push Pull	Pull-up *	Low	WD_SET0
	PG2	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	ROTRY_MUX_IN1
	PG3	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	ROTRY_MUX_IN2
	PG4	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	ROTRY_MUX_IN3
	PG5	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	ROTRY_MUX_IN4
	PG6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG3_4
	PG7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG1_LE
	PG8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG2_LE
	PC6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	READY_LED_IO2
	PC7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ABORT_LED_IO2
	PA15 (JTDI)	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SPI3_CS
	PD5	GPIO_EXTI5	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	DISCRETE_CH3_STS
	PD7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RS422_1_DE_RE
	PG11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RS422_LOOP_IN1
	PG12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RS422_LOOP_IN2
	PB4 (NJTRST)	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SEG3_BC
	PB8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ABORT_LED_IO1
	PB9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	READY_LED_IO1
	PE1	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	DIS_CH2_IN

3.2. DMA configuration

nothing configured in DMA service

3.3. NVIC configuration

3.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
USART1 global interrupt	true	0	0
USART2 global interrupt	true	0	0
TIM5 global interrupt	true	0	0
USB OTG FS global interrupt	true	0	0
PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/35/36/37/38	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
EXTI line2 interrupt	unused		
EXTI line4 interrupt	unused		
ADC1 and ADC2 interrupts	unused		
EXTI line[9:5] interrupts	unused		
TIM1 break interrupt and TIM15 global interrupt	unused		
TIM1 update interrupt and TIM16 global interrupt	unused		
TIM1 trigger and commutation interrupts and TIM17 global interrupt	unused		
TIM1 capture compare interrupt	unused		
TIM2 global interrupt	unused		
TIM3 global interrupt	unused		
TIM4 global interrupt	unused		
SPI1 global interrupt	unused		
SPI2 global interrupt	unused		
SPI3 global interrupt	unused		
FPU global interrupt	unused		

3.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
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Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true
USART1 global interrupt	false	true	true
USART2 global interrupt	false	true	true
TIM5 global interrupt	false	true	true
USB OTG FS global interrupt	false	true	true

*** User modified value**

4. System Views

4.1. Category view

4.1.1. Current

Middleware						
USB_DEVICE ✓						
System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing
DMA	ADC1 ✓	TIM1 ✓	SPI1 ✓			
GPIO ✓		TIM2 ✓	SPI2 ✓			
IWDG ✓		TIM3 ✓	SPI3 ✓			
IVVIC ✓		TIM4 ✓	USART1 ✓			
RCC ✓		TIM5 ✓	USART2 ✓			
SYS ✓			USB_FS ✓			

5. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32l4_bsd1.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32l4_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32l4_svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32l4-series-product-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/brstm32ulp.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32l4.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32gui.pdf
Magazine Articles	https://www.st.com/resource/en/magazine/design-elektronik_august2017.pdf
Magazine Articles	https://www.st.com/resource/en/magazine/design-elektronik_october2016.pdf
Product Certifications	https://www.st.com/resource/en/certification_document/sesip-2000002-01-cert.pdf
Product Certifications	https://www.st.com/resource/en/certification_document/sesip-2000002-01-st2.pdf

Product	https://www.st.com/resource/en/certification_document/psa-certificate_stm32l4.pdf
Certifications	
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-uart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4555-getting-started-with-stm32l4-series-and-stm32l4-series-hardware-development-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4612-migrating-from-stm32l1-series-to-stm32l4-series-and-stm32l4-series-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4616-migrating-from-stm32f401-and-stm32f411-lines-to-stm32l4-series-and-stm32l4-series-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4621-stm32l4-and-stm32l4-ultralowpower-features-overview-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4629-adc-hardware-oversampling-for-microcontrollers-of-the-stm32-l0-and-l4-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4649-migrating-from-

stm32f1-series-to-stm32l4-series--stm32l4-series-microcontrollers-
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Application Notes https://www.st.com/resource/en/application_note/an4726-stm32cube-firmware-examples-for-stm32l4-series-and-stm32l4-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4729-stm32l0l4-firewall-overview-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4730-using-the-firewall-embedded-in-stm32l0l4l4-series-mcus-for-secure-access-to-sensitive-parts-of-code-and-data-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-

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Application Notes https://www.st.com/resource/en/application_note/an2606-introduction-to-system-memory-boot-mode-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4044-floating-point-unit-demonstration-on-stm32-microcontrollers-stmicroelectronics.pdf

& Software

Application Notes https://www.st.com/resource/en/application_note/an4323-getting-started-with-stemwin-library-stmicroelectronics.pdf

& Software

Application Notes https://www.st.com/resource/en/application_note/an4435-guidelines-for-

for related Tools & Software	obtaining-ulcsaiec-607301603351-class-b-certification-in-any-stm32-application-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4631-how-to-calibrate-an-stm32l0xx-internal-rc-oscillator-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4657-stm32-inapplication-programming-iap-using-the-usart-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4666-parallel-synchronous-transmission-using-gpio-and-dma-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4701-proprietary-code-readout-protection-on-microcontrollers-of-the-stm32f4-series-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4726-stm32cube-firmware-examples-for-stm32l4-series-and-stm32l4-series-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4736-how-to-calibrate-stm32l4-series-microcontrollers-internal-rc-oscillator-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4758-proprietary-code-readout-protection-on-stm32l4-stm32l4-stm32g4-and-stm32wb-series-mcus-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4767-onthefly-firmware-update-for-dual-bank-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4834-implementation-of-transmitters-and-receivers-for-infrared-remote-control-protocols-with-stm32cube-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4841-digital-signal-processing-for-stm32-microcontrollers-using-cmsis-stmicroelectronics.pdf
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an4968-proprietary-code-read-out-protection-pcrop-on-stm32f72xxx-and-stm32f73xxx-

& Software	microcontrollers-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5056-integration-guide-for-the-xcubesbsfu-stm32cube-expansion-package-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5282-using-xcuberccalib-software-to-calibrate-stm32wb-series-internal-rc-oscillators-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5360-getting-started-with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5361-getting-started-with-projects-based-on-dualcore-stm32h7-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5418-how-to-build-a-simple-usbp-d-sink-application-with-stm32cubemx-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5426-migrating-graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-550-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dualcore-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4865-lowpower-timer-lptim-applicative-use-cases-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5698-adapting-the-xcubestl-functional-safety-package-for-stm32-iec-61508-compliant-to-other-safety-standards-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5731-stm32cubemx-and-stm32cubeide-threadsafe-solution-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5857-using-xcuberccalib-software-to-calibrate-stm32c0-series-internal-rc-oscillator-stmicroelectronics.pdf
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Application Notes https://www.st.com/resource/en/application_note/an4502-stm32-smbuspmbus-expansion-package-for-stm32cube-stmicroelectronics.pdf
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Application Notes https://www.st.com/resource/en/application_note/an5126-how-to-calibrate-internal-oscillators-on-stm32g0-mcus-stmicroelectronics.pdf
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& Software

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& Software

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